

UNIT 2

Extreme Values and Concavity

A Complete Coloured English Lesson

- Absolute and local extrema from graphs
- Extreme Value Theorem and the closed-interval method
- Critical numbers and the First Derivative Test
- Concavity, inflection points, and the Second Derivative Test
- Applications to position, velocity, acceleration, and optimisation
- Complete translated practice set from textbook pages 90–115

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Grade 12 Scientific Mathematics

Lesson Overview

Learning objectives

By the end of this lesson, students will be able to:

- identify local and absolute extrema from a graph;
- use the Extreme Value Theorem correctly;
- find absolute extrema of a continuous function on a closed interval;
- locate critical numbers and apply the First Derivative Test;
- determine intervals of concavity and locate inflection points;
- apply the Second Derivative Test when its hypotheses are satisfied;
- interpret velocity and acceleration using sign analysis.

Key vocabulary

Absolute maximum, absolute minimum, local maximum, local minimum, absolute extrema, local extrema, critical point, critical number, First Derivative Test, Second Derivative Test, concave up, concave down, inflection point.

The concentration of a medicine in a patient's blood, measured in $\mu\text{g/mL}$, is modelled by

$$C(t) = 3.59 + 8 \left(1.5e^{-0.4t-1} - e^{-0.6t} \right), \quad 0 \leq t \leq 12,$$

where t is measured in hours. Determine the time during the first 12 hours at which the concentration is greatest.

$$C'(t) = 4.8 \left(e^{-0.6t} - e^{-0.4t-1} \right).$$

Setting $C'(t) = 0$ gives $-0.6t = -0.4t - 1$, so $t = 5$. Comparing $C(0)$, $C(5)$, and $C(12)$ shows that the maximum concentration occurs after **5 hours**.

Absolute and Local Extreme Values

Absolute extrema

Let f have domain D , and let $c \in D$.

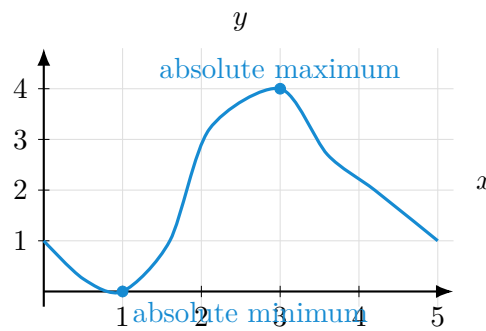
- $f(c)$ is an **absolute maximum value** of f on D if

$$f(c) \geq f(x) \quad \text{for every } x \in D.$$

- $f(c)$ is an **absolute minimum value** of f on D if

$$f(c) \leq f(x) \quad \text{for every } x \in D.$$

The absolute maximum and absolute minimum together are called the **absolute extreme values** of f .



For this graph on $[0, 5]$, $f(3) = 4$ is the absolute maximum value, and $f(1) = 0$ is the absolute minimum value.

Local extrema

Suppose c is an interior point of the domain of f .

- $f(c)$ is a **local maximum value** if $f(c) > f(x)$ for all nearby x in some open interval containing c .
- $f(c)$ is a **local minimum value** if $f(c) < f(x)$ for all nearby x in some open interval containing c .

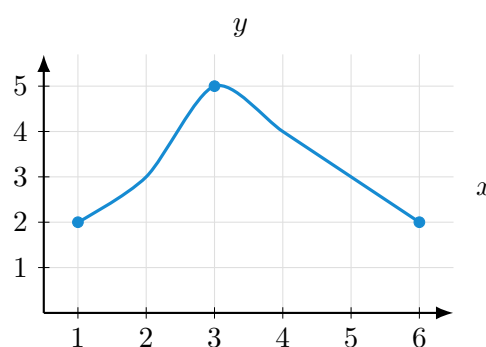
Local extrema compare $f(c)$ only with nearby values; absolute extrema compare it with every value in the domain.

Important endpoint distinction

An endpoint can be an absolute maximum or minimum, but under the textbook definition it is not a local extremum because it is not an interior point of the domain.

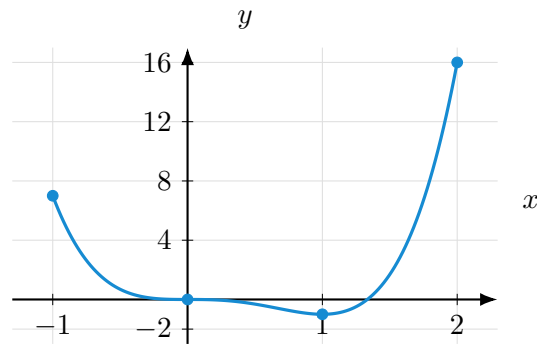
Example 1: Reading extrema from graphs

From the graph below, identify all local and absolute extrema.



Solution. The local and absolute maximum is $f(3) = 5$. The local and absolute minimum is $f(6) = 2$ in the textbook graph convention. The left endpoint $x = 1$ is not a local extremum.

For $f(x) = 3x^4 - 4x^3$ on $[-1, 2]$, the graph passes through $(-1, 7)$, $(0, 0)$, $(1, -1)$, and $(2, 16)$.



Solution. $f(1) = -1$ is a local and absolute minimum. $f(2) = 16$ is the absolute maximum, but it is not a local maximum because $x = 2$ is an endpoint. The critical value $x = 0$ does not produce an extremum.

For $f(x) = x^2$ on $\mathbb{R}$, $f(0) = 0$ is both a local minimum and an absolute minimum. The function has no maximum value.

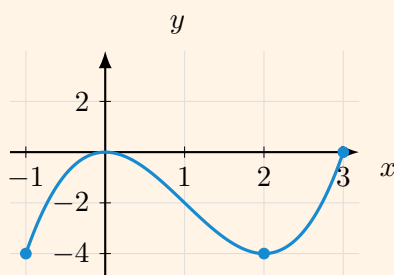
Consider $f(x) = x^2 + 1$ on $[-1, 2]$ with $x \neq 0$. The point $(0, 1)$ is missing.

- The absolute maximum is $f(2) = 5$.
- There is no absolute minimum because values of $f(x)$ approach 1 but never equal 1.

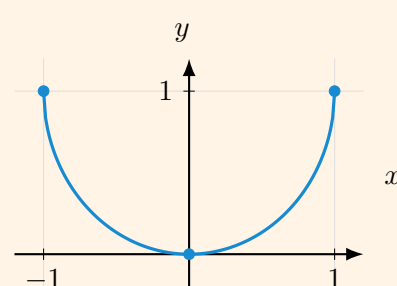
1. In Example 1A, is there an extremum at $x = 1$? Explain.
2. In Example 1B, is there an extremum at $x = 0$? Explain.
3. In Example 1D, why is 1 not an absolute minimum value?

Find all local and absolute extrema, if they exist.

(a) $f(x) = x^3 - 3x^2$ on $[-1, 3]$.



(b) $f(x) = 1 - \sqrt{1 - x^2}$ on $[-1, 1]$.



The Extreme Value Theorem

If f is continuous on a closed interval $[a, b]$, then f has both an absolute maximum value and an absolute minimum value on $[a, b]$.

What the theorem does and does not say

The theorem guarantees existence, not location. Absolute extrema may occur at interior critical numbers or at endpoints. If continuity or the closed interval condition is missing, absolute extrema may or may not exist.

Critical numbers and the closed-interval method

Critical numbers

An interior number c in the domain of f is a **critical number** if

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

The point $(c, f(c))$ is a critical point. Endpoints are handled separately in closed-interval problems and are not counted as critical numbers in this lesson. If f has a local extremum at c , then c must be a critical number. The converse is not always true.

To find the absolute extrema of a continuous function f on $[a, b]$:

1. Find every critical number of f in (a, b) .
2. Evaluate f at the critical numbers and at both endpoints.
3. The largest value is the absolute maximum; the smallest value is the absolute minimum.

Example 2: Absolute extrema on closed intervals

Find the absolute extrema of

$$f(x) = x^3 - 3x^2 - 9x + 2 \quad \text{on } [-2, 2].$$

Step 1: Critical numbers.

$$f'(x) = 3x^2 - 6x - 9 = 3(x + 1)(x - 3).$$

Only $x = -1$ lies in $(-2, 2)$.

Step 2: Evaluate candidates.

$$f(-2) = 0, \quad f(-1) = 7, \quad f(2) = -20.$$

Conclusion. The absolute maximum is 7 at $x = -1$, and the absolute minimum is -20 at $x = 2$.

Find the absolute extrema of $f(x) = x^{2/3}$ on $[-1, 2]$.

$$f'(x) = \frac{2}{3}x^{-1/3} = \frac{2}{3\sqrt[3]{x}}.$$

The derivative is undefined at $x = 0$, while $f(0)$ exists; hence $x = 0$ is critical. Evaluate:

$$f(-1) = 1, \quad f(0) = 0, \quad f(2) = \sqrt[3]{4}.$$

Thus the absolute maximum is $\sqrt[3]{4}$ at $x = 2$, and the absolute minimum is 0 at $x = 0$.

Find the absolute extrema of

$$f(x) = 2 \sin x - \cos 2x \quad \text{on } [0, 2\pi].$$

$$f'(x) = 2 \cos x + 2 \sin 2x = 2 \cos x(1 + 2 \sin x).$$

Therefore

$$x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}.$$

The candidate values are

x	0	$\frac{\pi}{2}$	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$	2π
$f(x)$	-1	3	$-\frac{3}{2}$	-1	$-\frac{3}{2}$	-1

Hence the absolute maximum is 3 at $x = \pi/2$, and the absolute minimum is $-3/2$ at $x = 7\pi/6$ and $x = 11\pi/6$.

Find the absolute maximum and minimum values, if they exist.

a. $f(x) = x^3 - 6x^2 + 5$ on $[-3, 5]$.

b. $f(x) = \sqrt[3]{x}$ on $[-8, 8]$.

c. $f(x) = \sin^2 x + \cos x$ on $[0, 2\pi]$.

The First Derivative Test

Increasing and decreasing intervals

If $f'(x) > 0$ throughout an interval I , then f is increasing on I . If $f'(x) < 0$ throughout I , then f is decreasing on I .

Suppose f is continuous at a critical number c .

- If f' changes from positive to negative at c , then $f(c)$ is a local maximum.
- If f' changes from negative to positive at c , then $f(c)$ is a local minimum.
- If f' has the same sign on both sides of c , then $f(c)$ is not a local extremum.

Find the local extrema of

$$f(x) = (x^2 - 3)e^x.$$

$$f'(x) = (x^2 + 2x - 3)e^x = (x - 1)(x + 3)e^x.$$

Because $e^x > 0$, the sign of f' is determined by $(x - 1)(x + 3)$:

interval	$(-\infty, -3)$	$(-3, 1)$	$(1, \infty)$
sign f'	+	-	+
f	increasing	decreasing	increasing

Therefore:

- local maximum at $x = -3$, with $f(-3) = 6e^{-3}$;
- local minimum at $x = 1$, with $f(1) = -2e$.

Find the local extrema, if any, of $f(x) = (x - 1)e^x$.

Find the local extrema of

$$f(x) = (x^2 - 4)^{2/3}.$$

$$f'(x) = \frac{4x}{3\sqrt[3]{x^2 - 4}}.$$

Critical numbers occur at $x = 0$ (where $f' = 0$) and $x = \pm 2$ (where f' is undefined but f exists). The sign chart is

interval	$(-\infty, -2)$	$(-2, 0)$	$(0, 2)$	$(2, \infty)$
sign f'	-	+	-	+
f	decreasing	increasing	decreasing	increasing

Thus:

- local minima at $x = -2$ and $x = 2$, both equal to 0;
- local maximum at $x = 0$, equal to $\sqrt[3]{16}$.

Find the local extrema, if any, of $f(x) = \sqrt[3]{x - 3}$.

Concavity and Inflection Points

Concavity

Let f be differentiable on an open interval I .

- The graph of f is **concave up** on I when f' is increasing on I .
- The graph of f is **concave down** on I when f' is decreasing on I .

Geometrically, a concave-up graph lies above its tangent lines, while a concave-down graph lies below its tangent lines.

If f'' exists on an open interval I , then:

- if $f''(x) > 0$ for all $x \in I$, the graph is concave up on I ;
- if $f''(x) < 0$ for all $x \in I$, the graph is concave down on I .

Inflection point

If f is continuous on an open interval containing c and the graph changes concavity at $x = c$, then $(c, f(c))$ is an **inflection point**.

If $(c, f(c))$ is an inflection point and $f''(c)$ exists, then $f''(c) = 0$. However, $f''(c) = 0$ or $f''(c)$ undefined does *not* by itself guarantee an inflection point; concavity must actually change, and the point must belong to the graph.

Example 5: Concavity and inflection points

Find the concavity intervals and inflection points of

$$f(x) = e^{-x^2/2}.$$

$$f'(x) = -xe^{-x^2/2}, \quad f''(x) = (x^2 - 1)e^{-x^2/2}.$$

Since $e^{-x^2/2} > 0$, the sign of f'' is determined by $x^2 - 1$.

interval	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
sign f''	+	-	+
concavity	up	down	up

Therefore, the graph is concave up on $(-\infty, -1) \cup (1, \infty)$ and concave down on $(-1, 1)$. The inflection points are

$$\left(-1, e^{-1/2}\right), \quad \left(1, e^{-1/2}\right).$$

Find the concavity intervals and inflection points of

$$f(x) = x + \frac{1}{x}, \quad x \neq 0.$$

$$f''(x) = \frac{2}{x^3}.$$

Thus the graph is concave down on $(-\infty, 0)$ and concave up on $(0, \infty)$. There is no inflection point at $x = 0$ because f is not defined there.

For $f(x) = x + 1/x$, determine all local and absolute extrema, if any.

Find intervals of concavity and all inflection points, if any.

a. $f(x) = (x - 2)^2(x - 1)$.

b. $f(x) = \frac{x}{x - 1}$.

The Second Derivative Test

Assume f' and f'' exist on an open interval containing c , and $f'(c) = 0$.

- If $f''(c) < 0$, then $f(c)$ is a local maximum.
- If $f''(c) > 0$, then $f(c)$ is a local minimum.
- If $f''(c) = 0$, the test is inconclusive; use the First Derivative Test or another method.

Use the Second Derivative Test to find the local extrema of

$$f(x) = (x^2 - 4)^2.$$

$$f'(x) = 4x(x^2 - 4),$$

so the critical numbers are $x = -2, 0, 2$. Also,

$$f''(x) = 12x^2 - 16.$$

Then

$$f''(-2) = 32 > 0, \quad f''(0) = -16 < 0, \quad f''(2) = 32 > 0.$$

Therefore:

- local minima: $f(-2) = 0$ and $f(2) = 0$;
- local maximum: $f(0) = 16$.

Use the Second Derivative Test to find the local extrema of $f(x) = xe^x$.

Application: Position, Velocity, and Acceleration

Motion relationships

If $s(t)$ is position, then

$$v(t) = s'(t), \quad a(t) = v'(t) = s''(t).$$

The object moves in the positive direction when $v(t) > 0$ and in the negative direction when $v(t) < 0$. Its velocity is increasing when $a(t) > 0$ and decreasing when $a(t) < 0$.

The position of an object is

$$s(t) = 3t^2 - 2t^3, \quad t \geq 0.$$

Direction of motion.

$$v(t) = 6t - 6t^2 = 6t(1 - t).$$

Thus $v(t) > 0$ on $(0, 1)$ and $v(t) < 0$ on $(1, \infty)$. The object moves in the positive direction on $(0, 1)$ and in the negative direction on $(1, \infty)$.

Increasing or decreasing velocity.

$$a(t) = 6 - 12t.$$

Thus $a(t) > 0$ on $(0, 1/2)$ and $a(t) < 0$ on $(1/2, \infty)$. Therefore velocity increases on $(0, 1/2)$ and decreases on $(1/2, \infty)$.

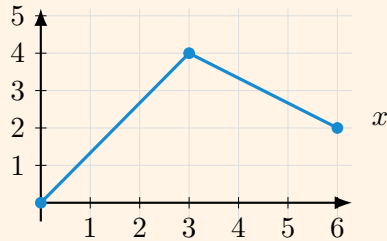
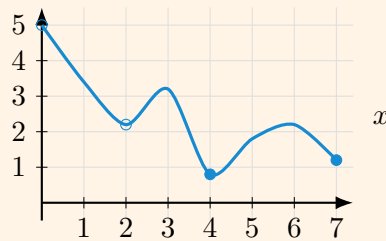
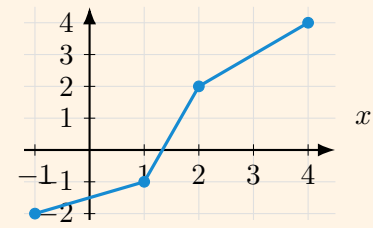
The position of an object is

$$s(t) = t^3 - 3t + 3, \quad t \geq 0.$$

- a. On which time intervals does the object move in the positive and negative directions?
- b. On which intervals is the velocity increasing or decreasing?

Complete Practice and Problem Solving

For each graph, find the critical numbers and all local and absolute extrema, if they exist.

1. $f(x)$ 2. $g(x)$ 3. $h(x)$ 

Find the absolute maximum and absolute minimum values, if they exist.

4. $f(x) = 1 + 6x - 3x^2$, $[0, 4]$.

7. $f(x) = \sqrt[3]{x}$, $[8, 64]$.

10. $f(x) = \frac{\ln x}{x^2}$, $[1/2, 4]$.

5. $f(x) = (x+3)^{2/3} - 5$, $[-3, 3]$.

8. $f(x) = 2\cos x + \sin 2x$, $[0, \pi/2]$.

11. $f(x) = \cos x$, $[-\pi/6, \pi/3]$.

6. $f(x) = \frac{x^2}{x^2 + 1}$, $[-2, 2]$.

9. $f(x) = \frac{e^x}{1 + x^2}$, $[0, 3]$.

12. $f(x) = \sqrt{4 - x^2}$, $[-2, 2]$.

Find intervals of increase and decrease, then find all local extrema.

13. $f(x) = x^3 - 6x^2 - 135$.

16. $f(x) = \sqrt{x^2 - 2x + 2}$.

14. $f(x) = \frac{2x}{x^2 + 9}$.

17. $f(x) = x^{2/3}(x - 3)$.

18. $f(x) = \sin^2 x + \sin x$, $[0, 2\pi]$.

15. $f(x) = x^2 \ln x$.

19. $f(x) = x + \sin x$, $[0, 2\pi]$.

Find intervals of concavity and all inflection points, if any.

20. $f(x) = x^3 - 12x + 1$.

23. $f(x) = \ln(x^2 + 5)$.

21. $f(x) = \sin x$, $[-\pi, \pi]$.

24. $f(x) = \sqrt{x}(x + 3)$.

22. $f(x) = \frac{3}{x^2 + 1}$.

25. $f(x) = xe^x$.

Find all local extrema. State when the Second Derivative Test is inconclusive or inapplicable.

26. $f(x) = 6x - x^2$.

29. $f(x) = x \ln x$.

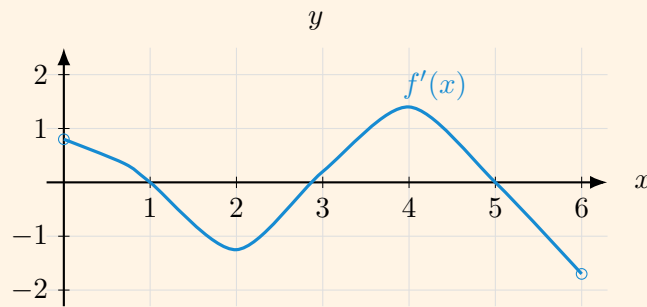
27. $f(x) = \cos x - x$, $[0, 4\pi]$.

30. $f(x) = \frac{x}{2^x}$.

28. $f(x) = \frac{x^2}{x - 1}$.

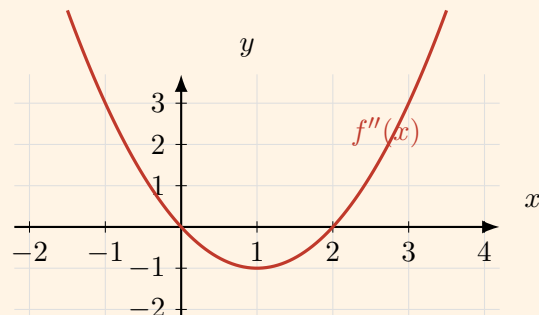
31. $f(x) = x^{2/3} - 3$.

The graph below represents $f'(x)$ for a continuous function f on $[0, 6]$.



32. Find the x -values at which f has local extrema, and classify each one.
33. Find the intervals on which f is increasing and decreasing.

34. Let $f(x) = x^3 + ax^2 + bx + c$. The function has a local maximum at $x = -3$ and a local minimum at $(1, -14)$. Find a , b , and c .
35. Let $f(x) = \sqrt{x+1} + \frac{b}{x}$. If the graph has an inflection point at $x = 3$, find b .



36. Find the intervals on which the graph of f is concave up and concave down.
37. Find the x -coordinates of the inflection points of f .

The graph represents the position $s(t)$, in metres, of an object moving on a straight line; t is in seconds.



38. Find the values of t at which the object is at rest.
39. On which intervals does the object move in the positive and negative directions?

40. If the acceleration is zero at $t = 4$, on which intervals is velocity increasing and decreasing?

A factory's weekly profit, in dinars, from selling x loudspeakers is

$$f(x) = \frac{1500}{x^2 - 6x + 10} - 150.$$

Find the number of loudspeakers that gives the greatest possible profit.

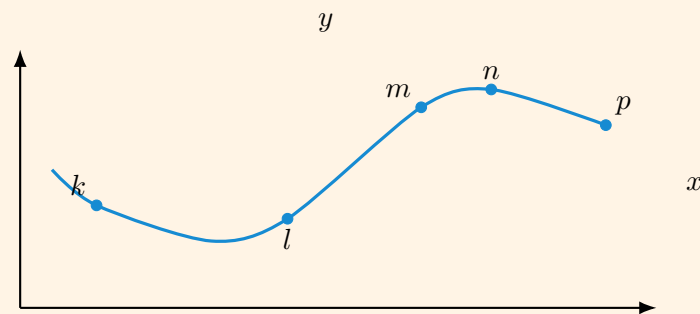
The position of an object is

$$s(t) = t^3 - 2t^2 + t, \quad t \geq 0.$$

42. On which intervals does the object move in the positive and negative directions?
 43. On which intervals is velocity increasing and decreasing?

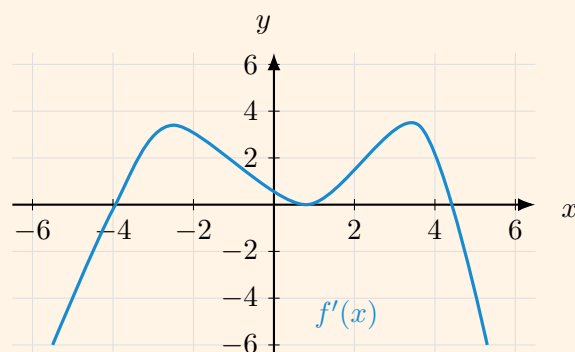
Higher-Order Thinking Skills

The labelled points lie on the graph of f .



Choose from $\{k, l, m, n, p\}$ and justify each answer.

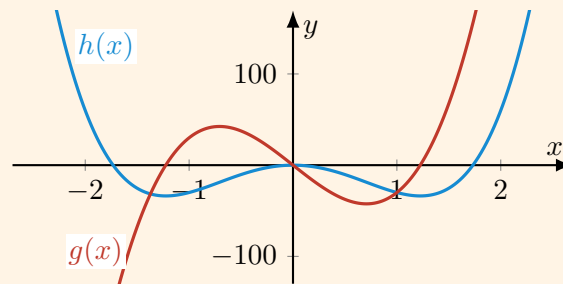
44. Both $f'(x)$ and $f''(x)$ are positive.
 45. Both $f'(x)$ and $f''(x)$ are negative.
 46. $f'(x)$ is negative and $f''(x)$ is positive.



47. Find the x -values where f has local extrema, and classify them.

48. Find the intervals on which f is increasing and decreasing.
49. Find the intervals on which f is concave up and concave down.
50. Find the x -coordinates of the inflection points of f .

The graph below shows two functions g and h . Determine which function is the derivative of the other, and justify your answer using zeros, signs, and turning points.



Lesson Summary

Core decision map

Absolute extrema on $[a, b]$	Critical numbers inside the interval plus both endpoints; compare function values.
Local extrema	Use a sign chart for f' or the Second Derivative Test when its conditions are satisfied.
Concavity	Use the sign of f'' .
Inflection point	Requires continuity and an actual change in concavity.
Motion direction	Use the sign of $v = s'$.
Increasing/decreasing velocity	Use the sign of $a = v' = s''$.

Final accuracy checklist

- A critical number must be in the domain of f .
- Endpoints are candidates for absolute extrema on a closed interval.
- $f'(c) = 0$ does not automatically imply an extremum.
- $f''(c) = 0$ does not automatically imply an inflection point.
- A change of concavity at a hole or vertical asymptote is not an inflection point.